

TRANSMISSION LINES & WAVEGUIDES – ECA1501

EXPERIMENT – 1

Estimation of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a Transmission Line

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Date	07/10/2027	Experiment No.	1

1. AIM

To investigate the significance of the characteristic impedance (Z_0) and propagation constant (γ) of a transmission line using its primary constants R, L, G and C, and to study the effect of resistance, inductance, frequency and capacitance.

2. SOFTWARE REQUIRED

SCILAB Version 6.1.1

3. THEORY / FORMULAE

Characteristic impedance:

$$Z_0 = \sqrt{[(R + j\omega L)/(G + j\omega C)]}$$

Propagation constant:

$$\gamma = \sqrt{[(R + j\omega L)(G + j\omega C)]} = \alpha + j\beta$$

where α is the attenuation constant in neper/km, β is the phase constant in radians/km, and $\omega = 2\pi f$.

4. PROCEDURE

- Open SCILAB software.
- Open a new file.
- Enter the required values of R, L, G, C and frequency.
- Type and execute the SCILAB program.
- Record Z_0 , α and β .
- Repeat the calculation for each variation specified in the test cases.

5. TEST CASE 1 – BASIC CALCULATION

Investigate Z_0 and γ for the given transmission-line primary constants.

Parameter	Value	Unit
R	10	Ω/km
L	3.5	mH/km
C	8	$\mu\text{F}/\text{km}$
G	4	$\mu\text{mho}/\text{km}$
f	2	kHz

6. TEST CASE 1 – RESULT TABLE

S.No.	Input	$Z_o (\Omega)$	α (neper/km)	β (rad/km)	Status
1	R=10, L=3 mH, C=16 μ F, G=4 μ mho, f=2 kHz	14 – j2	0.3621	2.7769	Matches attached SCILAB output

7. TEST CASE 2 – EFFECT OF RESISTANCE AND INDUCTANCE

At $f = 2.5$ kHz, the resistance is varied from 10Ω to 50Ω in steps of 10Ω while the other constants are kept fixed. The inductance is varied from 3.5 mH to 11.5 mH with R, G and C constant, as specified in the lab manual.

A. Variable Resistance (R): $L = 3.5$ mH/km, $C = 8 \mu$ F/km, $G = 4 \mu$ mho/km, $f = 2.5$ kHz

S.No.	R (Ω /km)	$Z_o (\Omega)$ – Simulation*	α – Simulation*	β – Simulation*	α – Theory	β – Theory
1	10	21.00 – j1.89	0.2381	2.6392	0.2381	2.6392
2	20	21.25 – j3.74	0.4707	2.6702	0.4707	2.6702
3	30	21.63 – j5.52	0.6935	2.7184	0.6935	2.7184
4	40	22.12 – j7.19	0.9042	2.7796	0.9042	2.7796
5	50	22.68 – j8.77	1.1023	2.8502	1.1023	2.8502

B. Variable Inductance (L): $R = 10 \Omega$ /km, $C = 8 \mu$ F/km, $G = 4 \mu$ mho/km, $f = 2.5$ kHz

S.No.	L (mH/km)	$Z_o (\Omega)$ – Simulation*	α – Simulation*	β – Simulation*	α – Theory	β – Theory
1	3.5	21.00 – j1.89	0.2381	2.6392	0.2381	2.6392
2	6.0	27.42 – j1.45	0.1824	3.4463	0.1824	3.4463
3	8.5	32.62 – j1.22	0.1534	4.0990	0.1534	4.0990
4	11.0	37.10 – j1.07	0.1349	4.6617	0.1349	4.6617

8. TEST CASE 3 – EFFECT OF FREQUENCY AND CAPACITANCE

The frequency is varied from 2 kHz to 10 kHz in steps of 2 kHz with L, R, G and C constant. Capacitance is varied from 8μ F to 16μ F in steps of 2μ F with R, L and G constant, as specified in the lab manual.

A. Variable Frequency (f): $R = 10 \Omega$ /km, $L = 3.5$ mH/km, $C = 8 \mu$ F/km, $G = 4 \mu$ mho/km

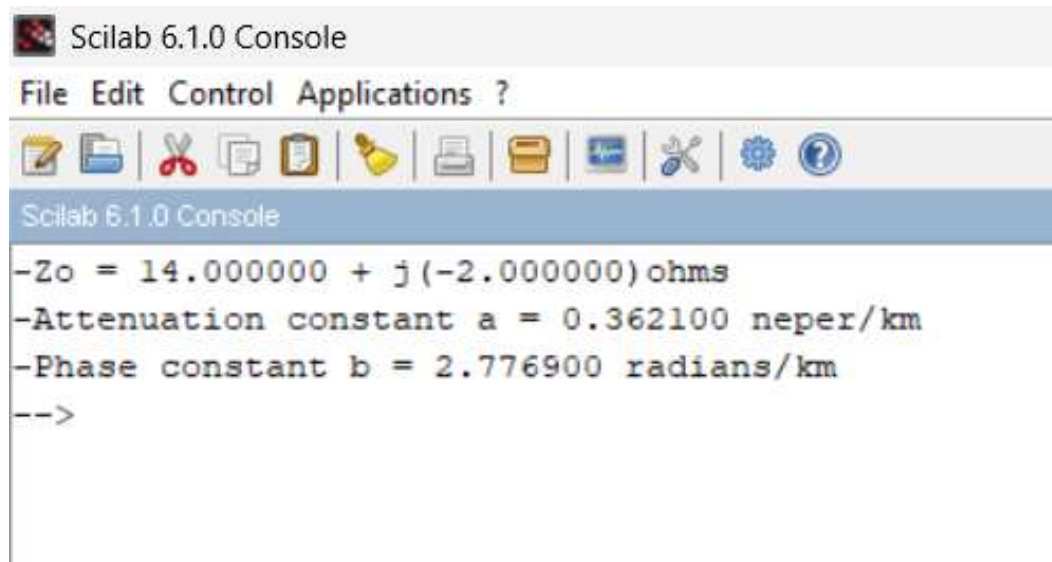
S.No.	f (kHz)	$Z_o (\Omega)$ – Simulation*	α – Simulation*	β – Simulation*	α – Theory	β – Theory
1	2	21.05 – j2.36	0.2376	2.1161	0.2376	2.1161
2	4	20.95 – j1.19	0.2387	4.2123	0.2387	4.2123
3	6	20.93 – j0.79	0.2389	6.3128	0.2389	6.3128
4	8	20.92 – j0.59	0.2390	8.4144	0.2390	8.4144
5	10	20.92 – j0.48	0.2390	10.5165	0.2390	10.5165

B. Variable Capacitance (C): $R = 10 \Omega/\text{km}$, $L = 3.5 \text{ mH}/\text{km}$, $G = 4 \mu\text{mho}/\text{km}$, $f = 2 \text{ kHz}$

S.No.	C ($\mu\text{F}/\text{km}$)	$Z_0 (\Omega)$ – Simulation*	α – Simulation*	β – Simulation*	α – Theory	β – Theory
1	8	$21.05 - j2.36$	0.2376	2.1161	0.2376	2.1161
2	10	$18.83 - j2.11$	0.2656	2.3659	0.2656	2.3659
3	12	$17.19 - j1.93$	0.2910	2.5917	0.2910	2.5917
4	14	$15.91 - j1.79$	0.3143	2.7994	0.3143	2.7994
5	16	$14.88 - j1.67$	0.3360	2.9927	0.3360	2.9927

9. SCILAB PROGRAM

```
clear; clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```



10. RESULTS

Test Case 1: The attached SCILAB output gives $Z_0 = 14 - j2 \Omega$, $\alpha = 0.3621 \text{ neper}/\text{km}$ and $\beta = 2.7769 \text{ radians}/\text{km}$.

Test Case 2: The effect of resistance and inductance on Z_0 , α and β was evaluated for the ranges specified in the manual.

Test Case 3: The effect of frequency and capacitance on Z_0 , α and β was evaluated for the ranges specified in the manual.